

K3 Surfaces from the Lattice Up

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ABSTRACT. A textbook on K3 surfaces built from the arithmetic of integral lattices upward, written for a reader who knows linear algebra, a first course in complex analysis, and the vocabulary of algebraic geometry, and who wants to reach the frontier where the transcendental lattice of a K3 surface, the monodromy of a Picard–Fuchs equation, and the geometry of a modular curve meet. Standard theorems are stated with references and marked **(L)**; exact computations that the reader can rerun are marked **(E)**; a small core of lattice facts that have been checked by the Lean 4 kernel in a public repository is marked **(K)**, and the last chapter explains what that mark does and does not mean. No physics is discussed.

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Preface

Why this book

Most introductions to K3 surfaces begin with the surface — a quartic in $\mathbb{P}^3$, a Kummer surface, a double plane branched along a sextic — and arrive at the lattice $H^2(X, \mathbb{Z})$ as a piece of topology one computes about it. This book runs the other way. It begins with integral lattices, because for K3 surfaces the lattice is not a shadow of the geometry: by the Torelli theorem it *is* the geometry, up to the position of a single line in a fixed 22-dimensional space. A reader who has internalized how a rank-3 lattice $U \oplus \langle 2n \rangle$ sits inside $U^3 \oplus E_8(-1)^2$, how a vector of norm -2 in it splits it, and how the isometries of that rank-3 lattice realize a modular group, already knows the skeleton of the theory of one-parameter families of lattice-polarized K3 surfaces. The rest is filling in the flesh — Hodge theory, periods, fibrations, differential equations — and knowing which classical theorem carries each step.

Who it is for

The reader we have in mind has taken linear algebra seriously (including bilinear forms over $\mathbb{Z}$), knows what a Riemann surface and a holomorphic form are, and has met the words “divisor”, “line bundle” and “cohomology” without necessarily being at ease with them. Chapters 1 and 7 require only the first of these. Chapters 2–6 use the rest, but every theorem is stated in full and every proof that we do not give is located precisely in a standard reference. We have tried to write the book we wished we had had when the mathematics of §8 was being assembled piece by piece from those references.

The status convention

Every mathematical statement in this book carries one of four marks.

- **(L)** *Literature*. A theorem of the published record, quoted with a precise citation. We do not reprove it; where a proof is short and illuminating we include it and say so.
- **(E)** *Exact computation*. A finite computation over $\mathbb{Q}$ or $\mathbb{Z}$ that the reader can rerun; typically a Gram matrix, a determinant, an order of vanishing, or a q -expansion checked to a stated order. We never write “verified” for a finite check without stating its order.
- **(N)** *Numerical*. A high-precision floating-point computation followed by recognition of an exact value. This is evidence, not proof, and the book says so wherever it uses one.
- **(K)** *Kernel-checked*. A statement that has been formalized in Lean 4 and accepted by its kernel, in a public repository, depending only on the three standard axioms of the library. Chapter 9 explains exactly what this mark certifies — and its two well-documented ways of certifying less than a name suggests.

A theorem marked **(L)** is not less true than one marked **(K)**. The marks record *how we know*, and a reader who wants to build on a statement should know which kind of knowing it is. The habit of writing the mark down is the single most useful thing this book has to teach.

How to read it

Chapter 1 is the foundation and should be read first; its worked examples are the lattices that return in every later chapter. Chapters 2–3 are the classical theory of K3 surfaces, told so that the lattice does the work. Chapter 4 is the one place where explicit equations appear, because elliptic fibrations are how one *computes* a Néron–Severi lattice by hand. Chapters 5

and 7 bring in analysis — Picard–Fuchs equations and modular curves — and Chapter 6 is where they meet the lattice at the points of Picard number 20. Chapter 8 assembles the frontier: what is known about Cooper’s sporadic sequences and the M_7 -polarized K3 family attached to them, with every claim carrying its mark. Chapter 9 is about the **(K)** mark itself.

Exercises are placed where the material is; hints are in Appendix A. An exercise marked $\star$ is one whose answer the book uses later.

What this book is not

It is not a research monograph: Huybrechts’ *Lectures on K3 Surfaces* [Huy16] and Barth–Hulek–Peters–Van de Ven [BHPVdV04] remain the references for proofs, and we lean on them constantly. It is not a treatise on formal verification; Chapter 9 is deliberately short. And it contains no physics: the lattice $U \oplus \langle 2n \rangle$ appears in string theory as well, and a reader from that direction will recognize some matrices, but every sentence here is a statement about mathematics and nothing else.

Provenance

This edition was drafted with the assistance of an AI model (Claude, Fable 5.1) working inside the repository that holds the Lean development of Chapter 9. The kernel-checked statements are transcribed from that repository at a stated commit; the literature statements are the author’s responsibility and have been checked against the cited sources to the extent stated in each chapter’s closing note. Corrections are welcome and will be recorded, not silently applied.

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CHAPTER 1

Integral lattices

Everything in this book that can be computed by hand is computed in a lattice. This chapter sets up the vocabulary — Gram matrices, discriminants, signatures, even and unimodular lattices, discriminant forms — and then does something introductory texts rarely do: it works, in complete detail, the handful of small lattices that carry the whole later theory. The reader who finishes the exercises will have met U , E_8 , $U \oplus \langle 2n \rangle$, the K3 lattice, and the binary forms A_2 , $\langle 2 \rangle \oplus \langle 2 \rangle$ and $\begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$ as concrete integer matrices, and will know exactly what it means to split one lattice off from another.

1. Definitions

DEFINITION 1.1. A *lattice* is a free $\mathbb{Z}$ -module L of finite rank r together with a symmetric bilinear form $\langle \cdot, \cdot \rangle: L \times L \rightarrow \mathbb{Z}$ that is non-degenerate, i.e. $\langle x, L \rangle = 0$ implies $x = 0$. Choosing a basis $(b_1, \dots, b_r)$ identifies L with $\mathbb{Z}^r$ and the form with its *Gram matrix* $G = (\langle b_i, b_j \rangle)$, a symmetric integer matrix with $\det G \neq 0$; then $\langle x, y \rangle = x^\top G y$. A change of basis by $P \in \mathrm{GL}_r(\mathbb{Z})$ replaces G by $P^\top G P$, so $\det G$ is an invariant of L , called the *discriminant* $\mathrm{disc} L$. The *norm* of x is $x^2 := \langle x, x \rangle$.

DEFINITION 1.2. L is *even* if $x^2 \in 2\mathbb{Z}$ for all x , equivalently if the diagonal of G is even; otherwise *odd*. L is *unimodular* if $\mathrm{disc} L = \pm 1$. The *signature* (s_+, s_-) counts the positive and negative eigenvalues of G over $\mathbb{R}$ (Sylvester); L is *definite* if one of them is zero, *indefinite* otherwise, and *hyperbolic* if $s_+ = 1$.

DEFINITION 1.3. For a sublattice $M \subseteq L$ of the same rank, the *index* $[L : M]$ is finite and $\mathrm{disc} M = [L : M]^2 \mathrm{disc} L$; if M is spanned by the columns of an integer matrix B in a basis of L , then $[L : M] = |\det B|$. A sublattice M of any rank is *primitive* if L/M is torsion-free, equivalently if $M = (M \otimes \mathbb{Q}) \cap L$. A vector v is primitive if $\mathbb{Z}v$ is.

DEFINITION 1.4. The *dual* of L is $L^* = \{x \in L \otimes \mathbb{Q} : \langle x, L \rangle \subseteq \mathbb{Z}\} \supseteq L$; in a basis, $L^* = G^{-1}\mathbb{Z}^r$. The finite group $A_L := L^*/L$ is the *discriminant group*, of order $|\mathrm{disc} L|$. For L even, $q_L: A_L \rightarrow \mathbb{Q}/2\mathbb{Z}$, $q_L(x + L) = x^2 \bmod 2\mathbb{Z}$, is well defined and is the *discriminant form*. For $x \in L$, the *divisor* $\mathrm{div}(x)$ is the positive generator of the ideal $\langle x, L \rangle \subseteq \mathbb{Z}$; $x/\mathrm{div}(x) \in L^*$, and x is primitive if and only if $x/\mathrm{div}(x)$ has order exactly $\mathrm{div}(x)$ in A_L .

EXAMPLE 1.5 (the hyperbolic plane). $U = \mathbb{Z}e \oplus \mathbb{Z}f$ with $e^2 = f^2 = 0$, $\langle e, f \rangle = 1$, Gram matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Even, unimodular, signature $(1, 1)$. The norm form is $2xy$; U is the unique even unimodular lattice of signature $(1, 1)$ **(L)** [Huy16, Ch. 14].

EXAMPLE 1.6 (the rank-one lattices). $\langle m \rangle$ denotes $\mathbb{Z}$ with Gram matrix (m) . It is even iff m is even, and $A_{\langle m \rangle} \cong \mathbb{Z}/m$ with $q(1) = 1/m \bmod 2$ when m is even.

EXAMPLE 1.7 (E_8). The E_8 lattice is the unique positive-definite even unimodular lattice of rank 8 **(L)** [CS99, Ch. 4]; a Gram matrix is the Cartan matrix of the root system E_8 . Its negative $E_8(-1)$, with Gram matrix negated, is negative-definite even unimodular. We shall never need its entries, only its two properties: even and unimodular.

THEOREM 1.8 **((L), even unimodular lattices)**. *If L is even and unimodular of signature (s_+, s_-) , then $s_+ - s_- \equiv 0 \pmod{8}$. Conversely, for every such indefinite signature there is exactly one even unimodular lattice, namely $U^{s_-} \oplus E_8^{(s_+ - s_-)/8}$ when $s_+ \geq s_-$, and analogously with $E_8(-1)$ otherwise [Ser73, Ch. V], [Huy16, Thm. 14.1.1].*

DEFINITION 1.9 (the K3 lattice). $\Lambda_{K3} := U^{\oplus 3} \oplus E_8(-1)^{\oplus 2}$, even, unimodular, of signature (3, 19) and rank 22. By Theorem 1.8 it is the unique even unimodular lattice of that signature.

2. Orthogonal complements and splittings

Let $v \in L$ with $v^2 \neq 0$. Its orthogonal complement $v^\perp = \{x \in L : \langle x, v \rangle = 0\}$ is a primitive sublattice of rank $r - 1$, and $\mathbb{Z}v \oplus v^\perp$ has full rank in L . How far is it from all of L ?

PROPOSITION 2.1 ((L); elementary). *Let $v \in L$ be primitive with $v^2 \neq 0$, $d = \text{div}(v)$. Then*

$$[L : \mathbb{Z}v \oplus v^\perp] = \frac{|v^2|}{d}, \quad \text{disc}(v^\perp) \cdot v^2 = \text{disc}(L) \cdot \left(\frac{v^2}{d}\right)^2.$$

PROOF. The map $L \rightarrow \mathbb{Z}$, $x \mapsto \langle x, v \rangle$, has image $d\mathbb{Z}$ and kernel $v^\perp$, so $L/v^\perp \cong \mathbb{Z}$ and $L/(\mathbb{Z}v \oplus v^\perp) \cong d\mathbb{Z}/\langle v, v \rangle\mathbb{Z}$, of order $|v^2|/d$. The second identity is $\text{disc}(\mathbb{Z}v \oplus v^\perp) = [L : \mathbb{Z}v \oplus v^\perp]^2 \text{disc} L$. $\square$

In particular $\mathbb{Z}v \oplus v^\perp = L$ exactly when $\text{div}(v) = |v^2|$, i.e. when $v/v^2 \in L^*$. For a (-2) -vector (“root”) this says: a root splits L as a direct sum iff its divisor is 2; if its divisor is 1 the index is 2.

The most useful way to *exhibit* $v^\perp$ is not as a kernel but as a parametrized family, together with a 2×2 (or smaller) Gram matrix on a chosen basis and a *frame* — the $r \times r$ matrix whose columns are v followed by a basis of $v^\perp$ — whose determinant is the index of Proposition 2.1. The following worked examples are the ones that the rest of the book depends on. They are stated for the lattice

$$T_N := U \oplus \langle 2N \rangle, \quad \text{basis } (e, f, w), \quad \langle x, y \rangle = x_0y_1 + x_1y_0 + 2N x_2y_2, \quad \text{disc } T_N = -2N.$$

T_N is even, of signature (2, 1) for $N > 0$, with $A_{T_N} \cong \mathbb{Z}/2N$ and $q(1) = 1/2N$.

Kernel-checked (K)

Every equality in Examples 2.2–2.5 below is kernel-checked in `Agora/Geometry/MnLattice.lean` of the repository described in Chapter 9 (declarations `rootEF_*`, `root7_*`, `root7Inf_*`, `root10Inf_*`, `TN_splits_at_rootEF`, `T7_splits_at_root7`, `T7_splits_at_root7Inf`, `T10_splits_at_root10Inf`). The orthogonal complements are stated there as equivalences “ $x \perp v$ iff x is of the following form”, not merely as two vectors orthogonal to v . Chapter 9 explains why that distinction is the whole point.

EXAMPLE 2.2 (the Fricke root, every N). $v_1 = e - f = (1, -1, 0)$. Then $v_1^2 = -2$, $\langle x, v_1 \rangle = -x_0 + x_1$, so $\text{div}(v_1) = 1$ and

$$x \perp v_1 \iff x = (a, a, b) = a(e + f) + bw.$$

On the basis $(e + f, w)$ the form is $\text{diag}(2, 2N)$; so $v_1^\perp \cong \langle 2 \rangle \oplus \langle 2N \rangle$, of discriminant $4N$. The frame $(v_1, e + f, w)$ has determinant 2, and indeed $|v_1^2|/\text{div} = 2/1 = 2$. Check: $4N \cdot (-2) = -2N \cdot 2^2$.

EXAMPLE 2.3 (a second root in T_7). $v_2 = (2, -4, 1) = 2e - 4f + w$ in T_7 . Then $v_2^2 = 2 \cdot 2 \cdot (-4) + 14 = -2$, and $\langle x, v_2 \rangle = -4x_0 + 2x_1 + 14x_2 = 2(-2x_0 + x_1 + 7x_2)$, so $\text{div}(v_2) = 2$ and

$$x \perp v_2 \iff x = (a, 2a - 7b, b) = a(1, 2, 0) + b(0, -7, 1).$$

On $(1, 2, 0), (0, -7, 1)$ the Gram matrix is $\begin{pmatrix} 4 & -7 \\ -7 & 14 \end{pmatrix}$, of determinant 7; the unimodular change of basis $\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$ gives the reduced basis $(2, -3, 1), (1, 2, 0)$ with Gram matrix $\begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$, the binary form $x^2 + xy + 2y^2$ of discriminant -7 . The frame $(v_2, (2, -3, 1), (1, 2, 0))$ has determinant -1 . Since $\text{div}(v_2) = 2 = |v_2^2|$, Proposition 2.1 says the index is 1: we have found an *integral isometry*

$$U \oplus \langle 14 \rangle \cong \langle -2 \rangle \oplus \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}.$$

EXAMPLE 2.4 (a class of norm -42 in T_7). $v_3 = (14, -14, -5)$. Then $v_3^2 = -392 + 350 = -42$, $\langle x, v_3 \rangle = 14(-x_0 + x_1 - 5x_2)$, $\text{div}(v_3) = 14$, and

$$x \perp v_3 \iff x = (a, a + 5b, b) = a(1, 1, 0) + b(0, 5, 1).$$

Gram matrix on that basis $(\frac{2}{5}, \frac{5}{14})$, determinant 3; reduced basis $(1, 1, 0), (-2, 3, 1)$ with Gram matrix $(\frac{2}{1} \frac{1}{2})$, the root lattice A_2 . Frame determinant $3 = 42/14$. Check: $3 \cdot (-42) = -14 \cdot 3^2$.

EXAMPLE 2.5 (a class of norm -20 in T_{10}). $v_4 = (10, -10, -3)$ in T_{10} . Then $v_4^2 = -200 + 180 = -20$, $\langle x, v_4 \rangle = 10(-x_0 + x_1 - 6x_2)$, $\text{div}(v_4) = 10$, $x \perp v_4 \iff x = (a, a + 6b, b)$; reduced basis $(1, 1, 0), (-3, 3, 1)$ with Gram matrix $\text{diag}(2, 2) = \langle 2 \rangle \oplus \langle 2 \rangle$; frame determinant $2 = 20/10$.

REMARK 2.6. The four examples exhibit all three behaviours of Proposition 2.1 in one small lattice: index 2 (a root of divisor 1), index 1 (a root of divisor 2, a genuine direct summand), index 3. The discriminants of the complements — 28, 7, 3 in T_7 and 40, 4 in T_{10} — will reappear in Chapter 6 as the discriminants of transcendental lattices of K3 surfaces with Picard number 20, and in Chapter 7 as the discriminants of CM points on the modular curves $X_0(7)^+$ and $X_0(10)^+$. Nothing in this chapter knows that; it is the lattice arithmetic that the later chapters will be *fed into*.

3. Embeddings, gluing, and Nikulin's theorems

Two questions dominate the use of lattices in K3 theory: when does a lattice M embed primitively into Λ_{K3} , and when is a lattice determined by its rank, signature and discriminant form? Both were answered by Nikulin [Nik80]; we quote the two statements used most.

THEOREM 3.1 ((**L**), Nikulin). *Let L be an even indefinite lattice with $\text{rk } L \geq 2 + \ell(A_L)$, where $\ell(A_L)$ is the minimal number of generators of A_L . Then L is determined up to isometry by its signature and discriminant form, and the natural map $\text{O}(L) \rightarrow \text{O}(q_L)$ is surjective [Nik80, Thm. 1.14.2].*

THEOREM 3.2 ((**L**), Nikulin). *Let M be an even lattice of signature (m_+, m_-) and L an even unimodular lattice of signature (ℓ_+, ℓ_-) . If $\ell_+ > m_+$, $\ell_- > m_-$ and $\text{rk } L - \text{rk } M > \ell(A_M)$, then M embeds primitively into L , and the orthogonal complement $M^\perp$ has discriminant form $-q_M$ [Nik80, Cor. 1.12.3, Prop. 1.6.1].*

EXAMPLE 3.3 (gluing $\langle 2N \rangle \oplus \langle -2N \rangle$ inside U). In U , the vectors $e + Nf$ and $e - Nf$ have norms $2N$ and $-2N$ and are orthogonal; they span a sublattice of index $2N$ (frame determinant $\det(\frac{1}{N} \frac{1}{N}) = -2N$). Both are primitive. This is the elementary mechanism behind every statement of the form “ $\langle 2N \rangle$ and $\langle -2N \rangle$ are glued along $\mathbb{Z}/2N$ to make a unimodular lattice” (**K**) (glue_congruence, glue_det, glue_primitive).

EXAMPLE 3.4 (M_n and its complement). Let $M_n := U \oplus E_8(-1)^2 \oplus \langle -2n \rangle$, of signature $(1, 18)$, rank 19, $A_{M_n} \cong \mathbb{Z}/2n$. By Theorem 3.2 it embeds primitively into Λ_{K3} , and any complement has rank 3, signature $(2, 1)$ and discriminant form $-q_{M_n}$; by Theorem 3.1 (here $\text{rk} = 3 \geq 2 + 1$) that complement is unique:

$$(M_n)^\perp \cong U \oplus \langle 2n \rangle = T_n.$$

This is the lattice statement underlying Dolgachev's theory of M_n -polarized K3 surfaces (Chapter 3); it is a theorem of the literature (**L**) [Dol96, §7]. An explicit witness for $n = 7$ inside U^3 — integer matrices B, C with $B^\top U^3 B = T_7$, $C^\top U^3 C = U \oplus \langle -14 \rangle$, $B^\top U^3 C = 0$, the $E_8(-1)^2$ summand splitting off because it is unimodular — is kernel-checked (**K**) (Agora/Geometry/Embedding.lean). What is *not* kernel-checked there is primitivity of the images and the identification of each as the other's complement; that step is Theorem 3.2.

4. Isometries and reflections

For a vector v with $v^2 \neq 0$ and $2\langle x, v \rangle / v^2 \in \mathbb{Z}$ for all x — automatic for a root, $v^2 = -2$ — the *reflection*

$$s_v(x) = x - \frac{2\langle x, v \rangle}{v^2} v$$

is an integral isometry of L fixing $v^\perp$ pointwise and sending v to $-v$. Reflections in roots generate the Weyl group of L , and the walls $v^\perp \otimes \mathbb{R}$ cut the positive cone into chambers; for a K3 surface these chambers are where the ample classes live (Chapter 3).

EXAMPLE 4.1 (the swap). In T_N , the exchange $\sigma: e \leftrightarrow f$, $w \mapsto w$, is an isometry, an involution, of determinant -1 ; it is the reflection in the Fricke root $e - f$ (**K**) (`swap_isometry`, `swap_involution`). Its fixed lattice is $v_1^\perp = \langle 2 \rangle \oplus \langle 2N \rangle$ of Example 2.2. In Chapter 7 this same matrix will be the Fricke involution $\tau \mapsto -1/(N\tau)$.

5. Exercises

EXERCISE 5.1. Show that $\mathbb{Z}/2N$ with $q(1) = 1/2N$ and $\mathbb{Z}/2N$ with $q(1) = -1/2N$ are the discriminant forms of $\langle 2N \rangle$ and $\langle -2N \rangle$, and that they are isomorphic as finite quadratic forms iff -1 is a square modulo $4N$ (for even N) resp. modulo $2N$ (for odd N)... or rather: determine exactly for which $N \leq 12$ they are isomorphic. (Hint: Appendix A.)

EXERCISE 5.2 ($\star$). Prove the identity $\text{disc}(v^\perp) \cdot v^2 = \text{disc}(L) \cdot (v^2 / \text{div } v)^2$ of Proposition 2.1 directly from a frame, i.e. from $\det(F^\top G F) = \det(F)^2 \det G$ applied to the frame $F = (v \mid B)$ with B a basis of $v^\perp$.

EXERCISE 5.3. In T_7 , find all roots v (norm -2) with $|v_0|, |v_1|, |v_2| \leq 5$, compute $\text{div}(v)$ for each, and sort them into the two classes “index 2” and “index 1”. Which reduced binary forms occur as $v^\perp$?

EXERCISE 5.4 ($\star$). Show that in T_N every vector of the form $(N, -N, -k)$ has $\langle x, v \rangle = N(-x_0 + x_1 - 2kx_2)$, hence $\text{div}(v) = N$ when $\gcd(N, 2k) = N$ divides appropriately, and compute v^2 and $\text{disc}(v^\perp)$ as functions of N, k . Recover Examples 2.4 ($N = 7, k = 5$) and 2.5 ($N = 10, k = 3$).

EXERCISE 5.5. Prove that $U \oplus \langle 2N \rangle$ and the lattice with Gram matrix $\begin{pmatrix} 0 & 0 & -1 \\ 0 & 2N & 0 \\ -1 & 0 & 0 \end{pmatrix}$ are isometric, by an explicit $P \in \text{GL}_3(\mathbb{Z})$. (Both conventions occur in the literature; Chapter 8 records a case where the choice caused confusion between two teams.)

EXERCISE 5.6. The lattice $G_0(N)$ with Gram matrix $\begin{pmatrix} 0 & 0 & -2 \\ 0 & 1 & 0 \\ -2 & 0 & 0 \end{pmatrix} \cdot N$ -adjusted as in Chapter 8, of discriminant $-4N^2$, carries an action of $\Gamma_0(N)$ by symmetric squares. Show that no lattice of discriminant $-4N^2$ can be isometric to T_N unless $N \in \{0, \frac{1}{2}\}$. (This trivial observation is the content of a kernel-checked theorem, `no_isometry_GON_TN`, recorded because the two lattices were once conflated under one name.)

Chapter note

Definitions and Theorems 1.8, 3.1, 3.2 follow Huybrechts [Huy16, Ch. 14] and Nikulin [Nik80]; statements were checked against those sources. Proposition 2.1 is folklore; the proof given is complete. The numerical content of Examples 2.2–2.5 is kernel-checked as described; the reader is encouraged to redo each by hand, which takes ten minutes and is the best possible introduction to the subject.

CHAPTER 2

K3 surfaces

1. Definition and first examples

DEFINITION 1.1. A *K3 surface* is a compact connected complex surface X with trivial canonical bundle, $\omega_X \cong \mathcal{O}_X$, and $H^1(X, \mathcal{O}_X) = 0$.

The name honours Kummer, Kähler and Kodaira, and the mountain K2. Every K3 surface is Kähler **(L)** (Siu, [Huy16, Thm. 7.1.1]); the projective ones are the algebraic K3 surfaces.

EXAMPLE 1.2 (quartics). A smooth quartic surface $X \subset \mathbb{P}^3$ is K3: adjunction gives $\omega_X \cong \mathcal{O}_X(4-4) = \mathcal{O}_X$, and $H^1(\mathcal{O}_X) = 0$ from the exact sequence $0 \rightarrow \mathcal{O}_{\mathbb{P}^3}(-4) \rightarrow \mathcal{O}_{\mathbb{P}^3} \rightarrow \mathcal{O}_X \rightarrow 0$. The Fermat quartic $x^4 + y^4 + z^4 + w^4 = 0$ is the standard example; its Picard number is 20, the maximum (Chapter 6).

EXAMPLE 1.3 (double planes). The double cover of $\mathbb{P}^2$ branched along a smooth sextic is K3. This is the generic member of the two-dimensional family with $\text{NS} \cong \langle 2 \rangle$.

EXAMPLE 1.4 (Kummer surfaces). Let A be a complex torus of dimension 2 and $\iota = -1$. The quotient A/ι has 16 ordinary double points; its minimal resolution $\text{Km}(A)$ is a K3 surface, the *Kummer surface* of A . The sixteen exceptional curves are disjoint (-2) -curves whose classes span, together with a certain half-sum, the *Kummer lattice* $K \subset \text{NS}(\text{Km } A)$ of rank 16 and discriminant 2^6 **(L)** [Huy16, Ch. 14.3]. Kummer surfaces of products $E_1 \times E_2$ of elliptic curves are the most computable K3 surfaces of all, and return in Chapters 4 and 6.

EXAMPLE 1.5 (elliptic K3 surfaces). A Weierstrass equation $y^2 = x^3 + a_4(t)x + a_6(t)$ with $\deg a_4 \leq 8$, $\deg a_6 \leq 12$ (and not both too small) defines, after minimal resolution, an elliptic K3 surface $X \rightarrow \mathbb{P}^1$. Chapter 4 is devoted to these.

2. The Hodge diamond and the lattice H^2

THEOREM 2.1 ((L)). For a K3 surface X : $b_1 = 0$, $b_2 = 22$, the Euler number is $e(X) = 24$, and the Hodge numbers are $h^{2,0} = h^{0,2} = 1$, $h^{1,1} = 20$. The intersection form makes $H^2(X, \mathbb{Z})$ an even unimodular lattice of signature $(3, 19)$, hence

$$H^2(X, \mathbb{Z}) \cong \Lambda_{K3} = U^3 \oplus E_8(-1)^2$$

by Theorem 1.8 [Huy16, Ch. 1, Prop. 3.5], [BHPVdV04, VIII.3].

Evenness is Wu's formula ($x^2 \equiv \langle x, w_2 \rangle$ and $w_2 = c_1 \bmod 2 = 0$), unimodularity is Poincaré duality, and the signature is the Hodge index theorem together with $b_2^+ = 2h^{2,0} + 1 = 3$. All K3 surfaces thus share *one* lattice. The whole theory is about which line inside it is $H^{2,0}$.

3. Néron–Severi and transcendental lattices

DEFINITION 3.1. $\text{NS}(X) := H^{1,1}(X) \cap H^2(X, \mathbb{Z})$ is the *Néron–Severi lattice*, a primitive sublattice of $H^2(X, \mathbb{Z})$; its rank $\rho(X)$ is the *Picard number*. The *transcendental lattice* is $T(X) := \text{NS}(X)^\perp$.

For a K3 surface $\text{Pic}(X) \cong \text{NS}(X)$ since $H^1(\mathcal{O}_X) = 0$, and by the Lefschetz $(1, 1)$ theorem $\text{NS}(X)$ is exactly the set of integral classes orthogonal to the holomorphic 2-form:

$$\text{NS}(X) = \{x \in H^2(X, \mathbb{Z}) : \langle x, \sigma \rangle = 0\}, \quad 0 \neq \sigma \in H^{2,0}(X).$$

Hence $T(X)$ is the smallest primitive sublattice whose complexification contains σ . Signatures: if X is projective, $\text{NS}(X)$ has signature $(1, \rho - 1)$ and $T(X)$ has signature $(2, 20 - \rho)$; in particular $\rho \leq 20$, and since $T \otimes \mathbb{C} \supseteq H^{2,0} \oplus H^{0,2}$ we have $\text{rk } T \geq 2$.

THEOREM 3.2 ((L), minimality of T). $T(X)_{\mathbb{Q}}$ is the smallest $\mathbb{Q}$ -sub-Hodge structure of $H^2(X, \mathbb{Q})$ containing $H^{2,0}$; equivalently, $T(X)_{\mathbb{Q}}$ is an irreducible Hodge structure [Huy16, Lem. 3.3.1]. If X is projective, the Hodge structure $T(X)_{\mathbb{Q}}$ is simple in the strong sense of Zarhin [Zar83, Thm. 1.6]: its endomorphism algebra is a field, totally real or CM.

COROLLARY 3.3. If a one-parameter family of projective K3 surfaces has generic Picard number 19, then the generic transcendental lattice has rank 3 and signature $(2, 1)$, and the Picard–Fuchs equation of a period has order 3 (Chapter 5).

WARNING 3.4. “ $\text{rk } T = 22 - \rho$ ” is arithmetic, not evidence. If one *assumes* a family is M_7 -polarized, then $\rho \geq 19$ generically and $\text{rk } T \leq 3$; the statement $\text{rk } T = 3$ then says nothing beyond the assumption. A recurring error in this subject is to present the number 3 as corroboration of the assumption that produced it. We shall flag the places where the temptation arises.

4. Picard number 20

DEFINITION 4.1. A K3 surface with $\rho(X) = 20$ is called *singular* (an unfortunate but standard term; the surface is smooth). Then $T(X)$ is a positive-definite even lattice of rank 2, i.e. a positive-definite even binary quadratic form $\begin{pmatrix} 2a & b \\ b & 2c \end{pmatrix}$.

THEOREM 4.2 ((L), Shioda–Inose). The map $X \mapsto T(X)$ is a bijection between singular K3 surfaces up to isomorphism and positive-definite even binary forms up to $\text{SL}_2(\mathbb{Z})$ equivalence [SI77]. Every singular K3 surface is defined over a number field, and its transcendental lattice determines it.

The forms $A_2 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $\langle 2 \rangle \oplus \langle 2 \rangle$ and $\begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$ of Chapter 1 are therefore the transcendental lattices of three specific K3 surfaces: the first two are the Kummer-related surfaces X_3 and X_4 of Shioda–Inose, of discriminants 3 and 4, the smallest possible; the third has discriminant 7. Chapter 6 explains how they are constructed from products of elliptic curves with complex multiplication.

5. The ample cone and (-2) -curves

For projective X , the positive cone $\{x \in \text{NS}(X)_{\mathbb{R}} : x^2 > 0\}$ has two components; call $\mathcal{C}_X$ the one containing an ample class. The *ample cone* is the set of $x \in \mathcal{C}_X$ with $\langle x, C \rangle > 0$ for every (-2) -curve C (a smooth rational curve; $C^2 = -2$ by adjunction). Equivalently (L) [Huy16, Cor. 8.1.6]: the ample cone is the chamber of $\mathcal{C}_X$ cut out by the walls $\delta^\perp$, $\delta \in \text{NS}(X)$, $\delta^2 = -2$, that contains an ample class. This is the geometric meaning of the reflections and walls of §4.1: the Weyl group of $\text{NS}(X)$ acts on the chambers, and every (-2) -class δ is, up to sign, effective (Riemann–Roch gives $h^0(\delta) + h^0(-\delta) \geq 2$).

6. Exercises

EXERCISE 6.1. Verify $e(X) = 24$ for a smooth quartic from c_2 of the tangent bundle, and for $\text{Km}(A)$ from $e(A) = 0$, the 16 fixed points, and the 16 exceptional $\mathbb{P}^1$'s.

EXERCISE 6.2 ($\star$). Let $\text{NS}(X) \supseteq U \oplus E_8(-1)^2 \oplus \langle -2n \rangle = M_n$ primitively, with $\rho = 19$. Show that $T(X) \cong U \oplus \langle 2n \rangle$ using Example 3.4, and identify the exact hypothesis being used (primitivity, and equality of NS with M_n).

EXERCISE 6.3. Show that a positive-definite even binary form of discriminant 3 is $\text{SL}_2(\mathbb{Z})$ -equivalent to A_2 , and one of discriminant 4 to $\langle 2 \rangle \oplus \langle 2 \rangle$. List the reduced forms of discriminants 7, 28, 40.

Chapter note

All theorems in this chapter are from Huybrechts [**Huy16**] and BHPV [**BHPVdV04**], and Shioda–Inose [**SI77**]; theorem and chapter numbers were checked against the cited editions to the extent the author had them at hand, and are given so the reader can go straight to the proof. The warning 3.4 records a lesson learnt in the program that produced this book (Chapter 8).

CHAPTER 3

Periods, Torelli, and lattice polarizations

The theme of this chapter is a single sentence: *a K3 surface is determined by the position of the line $H^{2,0}$ inside the fixed lattice Λ_{K3} , and every allowable position occurs*. Its two halves are the Torelli theorem and the surjectivity of the period map. Dolgachev’s theory of lattice-polarized K3 surfaces is then the systematic way to cut out the families one actually studies, and it produces the rank-3 lattice $T_n = U \oplus \langle 2n \rangle$ of Chapter 1 as the transcendental lattice of a one-parameter family whose parameter space is a modular curve.

1. The period domain

Fix a *marking*, an isometry $\phi: H^2(X, \mathbb{Z}) \rightarrow \Lambda := \Lambda_{K3}$. The line $\phi(H^{2,0}(X)) = \mathbb{C}\sigma$ satisfies the Riemann bilinear relations

$$\langle \sigma, \sigma \rangle = 0, \quad \langle \sigma, \bar{\sigma} \rangle > 0,$$

so it is a point of the *period domain*

$$\Omega := \{[\sigma] \in \mathbb{P}(\Lambda_{\mathbb{C}}) : \sigma^2 = 0, \sigma\bar{\sigma} > 0\},$$

an open subset of a 20-dimensional quadric, with two connected components exchanged by conjugation. The holomorphic 2-form is the only datum in H^2 that varies with the surface.

THEOREM 1.1 ((L), Local Torelli). *The period map from the base of a local universal deformation of X to Ω is a local isomorphism. In particular K3 surfaces have 20 moduli* [Huy16, Ch. 6, Prop. 2.8].

THEOREM 1.2 ((L), Global Torelli). *Two K3 surfaces X, X' are isomorphic if and only if there is a Hodge isometry $H^2(X, \mathbb{Z}) \cong H^2(X', \mathbb{Z})$. Moreover, a Hodge isometry that maps an ample (or Kähler) class to an ample (Kähler) class is induced by a unique isomorphism $X' \rightarrow X$* [PSS71], [BR75], [Huy16, Thm. 7.5.3].

THEOREM 1.3 ((L), surjectivity of the period map). *Every point of Ω is the period of some marked K3 surface* [Tod80, Loo81], [Huy16, Thm. 7.4.1].

Torelli is the statement that the lattice remembers everything; surjectivity, that the lattice forgets nothing. Together they turn questions about K3 surfaces into questions about lines in $\Lambda_{\mathbb{C}}$ and sublattices of Λ — which is why Chapter 1 came first.

EXAMPLE 1.4 (Picard number from the period). Given $[\sigma] \in \Omega$, the Néron–Severi lattice of the corresponding surface is $\sigma^{\perp} \cap \Lambda$ (Lefschetz (1, 1)). A generic $[\sigma]$ has $\sigma^{\perp} \cap \Lambda = 0$, so the generic K3 surface has $\rho = 0$ and is not projective. Requiring $\rho \geq 1$ is one condition; requiring $\rho \geq 19$ is nineteen, leaving a one-dimensional family; and $\rho = 20$ is a countable set of points — the singular K3 surfaces of Theorem 4.2, with $T = \sigma^{\perp\perp} \cap \Lambda$ of rank 2.

2. Lattice-polarized K3 surfaces

DEFINITION 2.1 (Dolgachev [Dol96]). Let M be an even hyperbolic lattice (signature (1, rk M − 1)) admitting a primitive embedding $M \hookrightarrow \Lambda$. An *M -polarized K3 surface* is a pair (X, j) with $j: M \hookrightarrow \text{NS}(X)$ a primitive embedding such that $j(M)$ contains a pseudo-ample class (one in the closure of the ample cone, i.e. nef and big). Two such pairs are isomorphic if there is an isomorphism of surfaces compatible with the embeddings.

Fix a primitive embedding $M \subset \Lambda$ and let $T := M^\perp \subset \Lambda$, of signature $(2, 20 - \text{rk } M)$... more precisely $(2, 20 - \text{rk } M + 1 - 1) = (2, 19 - \text{rk } M + 1)$; for $\text{rk } M = 19$ this is $(2, 1)$. The period of an M -polarized surface lies in

$$\Omega_M := \{[\sigma] \in \Omega : \sigma \perp M\} = \{[\sigma] \in \mathbb{P}(T_{\mathbb{C}}) : \sigma^2 = 0, \sigma\bar{\sigma} > 0\},$$

a domain of dimension $\text{rk } T - 2 = 20 - \text{rk } M$.

THEOREM 2.2 ((L), Dolgachev). *The coarse moduli space of M -polarized K3 surfaces is $K_M = \Gamma_M \backslash \Omega_M$ where Γ_M is the group of isometries of Λ acting trivially on M (identified with a subgroup of $\text{O}(T)$). Its dimension is $20 - \text{rk } M$, and the period map is an isomorphism onto the complement of a locally finite union of divisors (the “ (-2) -walls”) [Dol96, Thm. 3.1, Prop. 3.3, Thm. 7.3].*

The walls are where the pseudo-ample condition fails: the period becomes orthogonal to a (-2) -vector $\delta \in T$, which is then an algebraic class not in M but orthogonal to it, and the surface acquires a rational curve that the polarization did not see. We shall meet exactly two such walls in the family of Chapter 8, and Chapter 1’s Examples 2.2 and 2.3 are their roots.

3. The M_n -polarized families

Take $M = M_n = U \oplus E_8(-1)^2 \oplus \langle -2n \rangle$, rank 19, so $T = T_n = U \oplus \langle 2n \rangle$ (Example 3.4) and K_{M_n} is a curve.

THEOREM 3.1 ((L), Dolgachev [Dol96], §7). (1) $(M_n)^\perp = U \oplus \langle 2n \rangle$, uniquely.
 (2) $\text{O}^+(U \oplus \langle 2n \rangle) / \{\pm 1\} \cong \Gamma_0(n)^+$, the extension of $\Gamma_0(n)$ by the Fricke involution, and $K_{M_n} \cong \Gamma_0(n)^+ \backslash \mathbb{H}$, the Fricke modular curve $X_0(n)^+$ (Chapter 7).
 (3) The (-2) -walls removed from the ample locus form an explicit countable set (Thm. 7.3 there).
 (4) K_{M_n} has a canonical coordinate, an integer-coefficient Hauptmodul when $X_0(n)^+$ has genus 0 (Thm. 7.5 there).

The identification in (ii) can be made completely explicit, and it is instructive to see the isometry group of a rank-3 lattice be a modular group. Write the period as

$$\omega(\tau) = e - n\tau^2 f + \tau w \in T_n \otimes \mathbb{C}, \quad \omega(\tau)^2 = 2 \cdot 1 \cdot (-n\tau^2) + 2n\tau^2 = 0,$$

so $\tau \mapsto [\omega(\tau)]$ maps $\mathbb{H}$ onto a component of Ω_{M_n} (the isotropic vectors of the rank-3 lattice form a conic, and τ is its rational parameter). Then one checks:

PROPOSITION 3.2 ((K) for the identities, (L) for the group-theoretic converse). *There is an explicit representation $\rho: \text{SL}_2(\mathbb{Z}) \rightarrow \text{GL}_3(\mathbb{Z})$ (a symmetric square in the basis (e, f, w)) such that for $\gamma = \begin{pmatrix} a & b \\ nc & d \end{pmatrix} \in \Gamma_0(n)$,*

$$\rho(\gamma)^T T_n \rho(\gamma) = T_n, \quad \rho(\gamma) \omega(\tau) = (nc\tau + d)^2 \omega(\gamma\tau),$$

the second identity holding as a polynomial identity in τ with no determinant hypothesis; the swap σ of Example 4.1 satisfies $\sigma \omega(\tau) = (-n\tau^2) \omega(-1/(n\tau))$; and the Atkin–Lehner elements W_Q for $Q \parallel n$ act by integer matrices in the same way. That these exhaust $\text{O}^+(T_n) / \pm 1$ is Dolgachev’s theorem, quoted.

The kernel-checked identities are in `Agora/Geometry/ModularAction.lean` (`rho_isometry`, `rho_mul`, `rho_mulVec_period`, `rhoQ_*`) and `MnLattice.lean` (`swap_is_fricke`); see Chapter 9. The reader should notice what they do and do not say: they exhibit the modular group *inside* the isometry group, with the automorphy factor $(nc\tau + d)^2$ of weight 2 appearing as the scalar by which a lattice isometry rescales the period. The equality of the two groups is not among them.

REMARK 3.3 (the two rank-3 lattices). There is a second rank-3 lattice on which $\Gamma_0(n)$ acts by symmetric squares: the lattice of binary quadratic forms $ax^2 + bxy + cy^2$ with $n \mid a$, with the discriminant form $b^2 - 4nac$ (up to normalization), of discriminant $-4n^2$. It is *not* isometric

to T_n , whose discriminant is $-2n$ (an isometry preserves the discriminant; $-4n^2 = -2n$ forces $n \in \{0, \frac{1}{2}\}$). Both carry $\Gamma_0(n)^+$ -actions, both have signature $(2, 1)$, and they have been conflated in at least one working document. Keep them apart: the transcendental lattice is T_n ; the form lattice is a different object that happens to share a group.

4. Exercises

EXERCISE 4.1. Show that Ω has exactly two connected components and that they are interchanged by $\sigma \mapsto \bar{\sigma}$. Show that for T of signature $(2, 1)$, Ω_M is two copies of the upper half-plane.

EXERCISE 4.2 ($\star$). Verify by hand that $\omega(\tau) = e - n\tau^2 f + \tau w$ is isotropic in T_n , that $\langle e - f, \omega(\tau) \rangle = -(n\tau^2 + 1)$, and hence that the wall of the Fricke root $e - f$ is the locus $n\tau^2 = -1$, i.e. $\tau = i/\sqrt{n}$ in $\mathbb{H}$. Compute the wall of the root $(2, -4, 1)$ in T_7 and show it is $\tau = \frac{1}{2} \pm i/\sqrt{28}$; compute the wall of $(-2, 4, 1)$ and show its roots differ from these by 1.

EXERCISE 4.3. Write down $\rho(\gamma)$ for $\gamma = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ (translation) and check the period identity directly.

Chapter note

Theorems 1.2–1.3 and the lattice-polarized theory are quoted from Huybrechts [Huy16] and Dolgachev [Dol96]; Dolgachev’s §7 was read from a hash-pinned copy in the program’s literature manifest. Proposition 3.2 states what the kernel checked, as transcribed from the repository at the commit of Chapter 9.

CHAPTER 4

Elliptic fibrations

An elliptic K3 surface is the one kind of K3 surface whose Néron–Severi lattice can be computed by hand from an equation. The tools are Kodaira’s classification of singular fibres, Tate’s algorithm for reading the fibre type off the Weierstrass coefficients, and the Shioda–Tate formula. We state them, then work the one family this book cares about — the Inose pencil — to the point where the reader can see the lattices A_2 and $\langle 2 \rangle \oplus \langle 2 \rangle$ of Chapter 1 emerge as transcendental lattices.

1. Weierstrass models

DEFINITION 1.1. An *elliptic surface* is a smooth projective surface X with a surjective morphism $\pi: X \rightarrow C$ to a curve, whose generic fibre is a smooth curve of genus 1, with no (-1) -curves in fibres (relative minimality). A *section* is a curve $O \subset X$ with $\pi|_O$ an isomorphism. If a section exists, the generic fibre is an elliptic curve over the function field of C and X has a *Weierstrass model*: over $C = \mathbb{P}^1$ with coordinate t ,

$$y^2 = x^3 + a_4(t)x + a_6(t), \quad a_4, a_6 \in \mathbb{C}[t],$$

with discriminant $\Delta = -16(4a_4^3 + 27a_6^2)$ and $j = 1728 \cdot 4a_4^3 / (4a_4^3 + 27a_6^2)$.

THEOREM 1.2 ((L)). For $C = \mathbb{P}^1$, the surface X is K3 if and only if, after making the model minimal, a_4 and a_6 have degrees ≤ 8 and ≤ 12 (as sections of $\mathcal{O}(8)$ and $\mathcal{O}(12)$), not both of lower “weight” than required; then Δ has degree 24, and the Euler number $e(X) = \sum_v e(F_v) = 24$ equals the sum of the orders of vanishing of Δ at the singular fibres (with ∞ included via the degree count) [Huy16, Ch. 11], [SS10, § 4].

2. Kodaira’s classification and Tate’s algorithm

THEOREM 2.1 ((L), Kodaira [Kod63]). A singular fibre of a relatively minimal elliptic surface is one of the types I_n ($n \geq 1$), II , III , IV , I_n^* ($n \geq 0$), IV^* , III^* , II^* (and, without a section, multiples mI_n). The Euler number of the fibre, the number m_v of its components, and the root lattice spanned by the components not meeting the zero section are:

type	$e(F_v)$	m_v	root lattice R_v	disc R_v
I_n	n	n	A_{n-1}	n
II	2	1	0	1
III	3	2	A_1	2
IV	4	3	A_2	3
I_n^*	$n+6$	$n+5$	D_{n+4}	4
IV^*	8	7	E_6	3
III^*	9	8	E_7	2
II^*	10	9	E_8	1

THEOREM 2.2 ((L), Tate’s algorithm, characteristic 0 [Tat75]). For a minimal Weierstrass model at a place $t = t_0$, the fibre type is read off the orders of vanishing ($\text{ord } a_4, \text{ord } a_6, \text{ord } \Delta$) as follows:

<i>type</i>	$\text{ord } a_4$	$\text{ord } a_6$	$\text{ord } \Delta$
I_0 (<i>smooth</i>)	≥ 0	≥ 0	0
I_n	0	0	n
II	≥ 1	1	2
III	1	≥ 2	3
IV	≥ 2	2	4
I_0^*	≥ 2	≥ 3	6
I_n^*	2	3	$n + 6$
IV^*	≥ 3	4	8
III^*	3	≥ 5	9
II^*	≥ 4	5	10

If $\text{ord } a_4 \geq 4$ and $\text{ord } a_6 \geq 6$ the model is not minimal at t_0 .

WARNING 2.3. The table is the one place in this book where a small memory slip has large consequences. An order-2 vanishing of Δ with a_4, a_6 nonvanishing is I_2 (an A_1), not I_1 ; an order-4 vanishing with $a_4 \equiv 0$ and $\text{ord } a_6 = 2$ is IV (A_2). Both distinctions decide Picard numbers below. When in doubt, compute the type from the singularity of the Weierstrass surface, not from memory — or check the Euler number sum.

3. The Shioda–Tate formula

THEOREM 3.1 ((L), Shioda–Tate [Shi90]). *Let $X \rightarrow \mathbb{P}^1$ be an elliptic surface with a section O and generic fibre F . Then*

$$\text{NS}(X) \supseteq U_{O,F} \oplus \bigoplus_v R_v(-1), \quad \rho(X) = 2 + \sum_v (m_v - 1) + \text{rk MW}(X),$$

where $U_{O,F}$ is the hyperbolic plane spanned by O and F (after the change of basis $O + F$), R_v are the fibre root lattices of Theorem 2.1, and $\text{MW}(X)$ is the Mordell–Weil group of sections. The sublattice $U \oplus \bigoplus_v R_v(-1)$ is the trivial lattice; it has finite index in $\text{NS}(X)$ when MW is finite, and $|\text{disc NS}(X)| = |\text{disc}(\text{trivial})|/|\text{MW}_{\text{tors}}|^2$ in that case.

Since $|\text{disc NS}(X)| = |\text{disc } T(X)|$ for a K3 surface (unimodularity of Λ), computing the trivial lattice and the torsion of MW computes $\text{disc } T(X)$ — and for $\rho = 20$, that together with positivity pins $T(X)$ down to a short list of binary forms.

4. The Inose pencil

DEFINITION 4.1 (Inose [Ino78]). For $(\alpha, \beta) \in \mathbb{C}^2$ the *Inose surface* $X_{\alpha,\beta}$ is the K3 surface with Weierstrass model

$$y^2 = x^3 - 3\alpha t^4 x + t^5 (t^2 - 2\beta t + 1).$$

THEOREM 4.2 ((L), Inose; Shioda [Shi06]). *For elliptic curves E_1, E_2 with normalized j -invariants $J_i = j(E_i)/1728$, choose α, β with $\alpha^3 = J_1 J_2$ and $\beta^2 = (1 - J_1)(1 - J_2)$. Then $X_{\alpha,\beta}$ carries a Shioda–Inose structure with $T(X_{\alpha,\beta}) \cong T(E_1 \times E_2)$; in particular $X_{\alpha,\beta}$ is a double cover of $\text{Km}(E_1 \times E_2)$ up to birational equivalence, and conversely every Kummer surface of a product arises this way.*

The pencil has fibres of type II^* at $t = 0$ and $t = \infty$ for all (α, β) : indeed $\text{ord}_0 a_6 = 5$, $\text{ord}_0 a_4 \geq 4$, $\text{ord}_0 \Delta = 10$, and at ∞ (weights 8, 12, 24) the same. That accounts for 20 of the 24 in the Euler number and for $E_8 \oplus E_8$ in the trivial lattice; the remaining four are distributed among the roots of $t^2 - 2\beta t + 1$ and the zeros of $4a_4^3 + 27a_6^2$ elsewhere.

EXAMPLE 4.3 ((E); the three special members). An exact computation of orders of vanishing (a polynomial computation over $\mathbb{Q}(i)$, rerunnable in any computer algebra system) gives:

(α, β)	finite places $t_0 \neq 0$: (ord a_4 , ord a_6 , ord Δ)	Euler sum
$(0, 1)$	$t = 1$: $(\infty, 2, 4)$	$10 + 10 + 4 = 24$
$(1, 0)$	$t = \pm 1$: $(0, 0, 2)$ each	$10 + 10 + 2 + 2 = 24$
$(0, 0)$	$t = \pm i$: $(\infty, 1, 2)$ each	$10 + 10 + 2 + 2 = 24$

Reading Theorem 2.2 (L): at $(0, 1)$ the fibre at $t = 1$ is $IV(A_2)$, so the trivial lattice is $U \oplus E_8(-1)^2 \oplus A_2(-1)$ of rank 20 and discriminant 3; at $(1, 0)$ the fibres at $t = \pm 1$ are $I_2(A_1)$ each, trivial lattice of rank 20 and discriminant 4; at $(0, 0)$ the fibres are II (no root), trivial lattice of rank 18. Since $\rho \leq 20$, the first two are singular K3 surfaces with $|\text{disc } T| = 3$ and 4 (if MW is torsion-free of rank 0, which one checks separately), hence by Theorem 4.2 and the classification of forms, $T(X_{0,1}) \cong A_2$ and $T(X_{1,0}) \cong \langle 2 \rangle \oplus \langle 2 \rangle$. The third member has $\rho \geq 18$ from fibres alone; Theorem 4.2 identifies it ($J_1 J_2 = 0$, $(1 - J_1)(1 - J_2) = 0$, so $\{J_1, J_2\} = \{0, 1\}$) as the Shioda–Inose partner of $E_\omega \times E_i$, two non-isogenous CM curves, for which $\rho = 18$ exactly.

The first two are, in Shioda–Inose’s notation, the surfaces X_3 and X_4 : $X_{0,1}$ corresponds to $J_1 = J_2 = 0$, i.e. $E_\omega \times E_\omega$ with $\omega = e^{2\pi i/3}$, and $X_{1,0}$ to $J_1 = J_2 = 1$, i.e. $E_i \times E_i$. We shall meet them again in Chapter 6 as the two points of smallest discriminant on two different modular curves.

REMARK 4.4 (a recorded slip). In the program from which this book grew, the point $(0, 0)$ was at one stage labelled “ X_4 ” and the fibres at $(1, 0)$ were at one stage labelled I_1 ; both errors were caught by the Euler-number and rank counts above and are recorded in the program’s briefs. The table is here so the reader does not repeat them.

5. Exercises

EXERCISE 5.1. Verify the entries of Theorem 2.1 for I_n and II^* from the dual graphs (a cycle of n rational curves; the affine $\tilde{E}_8$ diagram).

EXERCISE 5.2 ($\star$). Compute the orders of vanishing in Example 4.3 by hand for $(\alpha, \beta) = (0, 1)$: $a_4 = 0$, $a_6 = t^5(t - 1)^2$, $\Delta = -16 \cdot 27 t^{10}(t - 1)^4$.

EXERCISE 5.3. For $(\alpha, \beta) = (1, 0)$, show that $4a_4^3 + 27a_6^2 = 27t^{10}((t^2 + 1)^2 - 4t^2)$ and factor it; confirm $\text{ord}_{t=\pm 1} = 2$ and that $a_4(\pm 1) \neq 0$.

EXERCISE 5.4. Using Theorem 3.1, show that an elliptic K3 surface with two II^* fibres has $\rho \geq 18$, and that $\rho = 20$ requires either two more fibre components in total or Mordell–Weil rank 2.

Chapter note

Theorems 2.1, 2.2, 3.1 are classical; the tables were checked against Schütt–Shioda’s survey [SS10] and Silverman’s account of Tate’s algorithm [Sil94, IV.9]. Theorem 4.2 is Inose’s with Shioda’s normalization; the relations $\alpha^3 = J_1 J_2$, $\beta^2 = (1 - J_1)(1 - J_2)$ are as in [Shi06] and should be checked there before being relied on. The orders of vanishing in Example 4.3 are an exact computation performed for this book; the fibre labels are the table’s, and the identification of $(0, 1)$ and $(1, 0)$ with X_3, X_4 follows from Theorem 4.2. An independent computation in the program (Kuwata–Shioda’s model of the same pencil) gave the same orders $\{10, 10, 4\}$ at $J = 0$ and $\{10, 10, 2, 2\}$ at $J = 1$.

CHAPTER 5

Periods as functions: Picard–Fuchs equations

A one-parameter family of K3 surfaces X_z has a holomorphic 2-form σ_z varying with z , and its integrals over a basis of transcendental cycles satisfy a linear differential equation in z , the Picard–Fuchs equation. Its order is the rank of the transcendental lattice; for the M_n -polarized families that is 3, and Doran’s theorem says the equation is then always a *symmetric square* of a second-order one. This chapter explains that circle of ideas and connects it to the Apéry-like sequences of Zagier and Cooper, which is where the frontier of Chapter 8 lives.

1. Variation of Hodge structure and the Picard–Fuchs operator

Let $\pi: \mathcal{X} \rightarrow S$ be a smooth family of K3 surfaces over a curve S , and let $\mathcal{H} = R^2\pi_*\mathbb{Z}$ be the local system of second cohomology, with fibres Λ_{K3} . The Gauss–Manin connection ∇ differentiates sections. Let $\sigma \in \Gamma(S, \mathcal{H} \otimes \mathcal{O})$ be a family of holomorphic 2-forms; then $\sigma, \nabla\sigma, \nabla^2\sigma, \dots$ are sections of $\mathcal{T} \otimes \mathcal{O}$ where $\mathcal{T}$ is the local system of transcendental lattices (assuming ρ constant), of rank r , and they become linearly dependent at step r :

DEFINITION 1.1. The *Picard–Fuchs operator* of the family is the monic differential operator $L = \partial_z^r + c_{r-1}(z)\partial_z^{r-1} + \dots + c_0(z)$, with $c_i \in \mathbb{C}(z)$, such that $L\sigma = 0$; equivalently, the period functions $\int_\gamma \sigma_z$ for $\gamma \in \mathcal{T}^\vee$ span its solution space. Its order is $r = \text{rk } \mathcal{T}$.

In the θ -form, $\theta = z d/dz$, the operator is $L = \sum_i P_i(z)\theta^i$ with polynomial coefficients; this is the form in which Apéry-like recurrences arise, since $\theta z^n = n z^n$ turns $L \sum a_n z^n = 0$ into a recurrence for (a_n) .

THEOREM 1.2 ((L), Griffiths transversality; regular singularities). $\nabla\sigma \in \mathcal{F}^1 \otimes \mathcal{O}$, so $\nabla\sigma \perp \sigma$; the Picard–Fuchs equation is Fuchsian (regular singular points), with monodromy the image of $\pi_1(S)$ in $\text{O}(T)$ [Voi02, Ch. 10].

2. Doran’s theorem

THEOREM 2.1 ((L), Doran [Dor00], Thm. 5.13). The Picard–Fuchs equation of a family of M_n -polarized K3 surfaces is the symmetric square of a second-order homogeneous linear Fuchsian ordinary differential equation.

Two proofs are indicated in [Dor00]. The first is a general principle: the period point of an M_n -polarized surface lies on the quadric $\sigma^2 = 0$ in $\mathbb{P}(T_\mathbb{C}) \cong \mathbb{P}^2$, so the three period functions satisfy a nondegenerate quadratic relation; and a third-order equation whose solutions satisfy a nondegenerate quadratic relation is the symmetric square of a second-order one (Doran’s Cor. 5.8, which we quote). Explicitly, in the parametrization $\omega(\tau) = e - n\tau^2 f + \tau w$ of Chapter 3, the three coordinates $(1, -n\tau^2, \tau)$ are the monomials $1, \tau^2, \tau$ in the two solutions $1, \tau$ of a second-order equation — which is exactly what a symmetric square means. The second proof uses the Shioda–Inose presentation of the surfaces as coming from products of n -isogenous elliptic curves.

DEFINITION 2.2 (symmetric square). For $L_2 = \theta^2 + p_1\theta + p_0$ with solutions y_1, y_2 , the *symmetric square* $\text{Sym}^2 L_2$ is the third-order operator annihilating $y_1^2, y_1 y_2, y_2^2$. In θ -form with polynomial coefficients the coefficients of $\text{Sym}^2 L_2$ are explicit polynomials in those of L_2 and their θ -derivatives; the recipe is classical (see e.g. [AvS21, §4]).

WARNING 2.3. Doran’s theorem runs from geometry to analysis: *if* the family is M_n -polarized *then* its operator is a symmetric square. The converse — that an order-3 operator which happens to be a symmetric square comes from an M_n -polarized family — is not a theorem; Doran himself notes in [Dor00, §6] that the classification of rank-19 lattice polarizations needed for such a converse is lacking. Chapter 8 lives precisely in the gap.

3. Apéry-like sequences and their operators

EXAMPLE 3.1 (Apéry’s $\zeta(3)$ sequence). Apéry’s proof of the irrationality of $\zeta(3)$ uses $a_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$, satisfying

$$(n+1)^3 a_{n+1} - (34n^3 + 51n^2 + 27n + 5)a_n + n^3 a_{n-1} = 0.$$

Beukers and Peters [BP84] showed that $\sum a_n z^n$ is a period of a one-parameter family of K3 surfaces with generic Picard number 19 — the first appearance of Corollary 3.3 in the wild — and Peters–Stienstra identified the family as M_6 -polarized in today’s language.

THEOREM 3.2 ((L), Zagier’s sporadic sequences [Zag09]). *Among the second-order recurrences $(n+1)^2 u_{n+1} - (an^2 + an + b)u_n + cn^2 u_{n-1} = 0$ with $u_0 = 1$, $u_{-1} = 0$, exactly six “sporadic” parameter triples (a, b, c) (beyond degenerate and Legendrian ones) give integer sequences; each $\sum u_n z^n$ is a modular form of weight 1 in a Hauptmodul of a genus-zero group of level ≤ 8 or so.*

THEOREM 3.3 ((L), Cooper [Coo12]; Gorodetsky [Gor21]). *There are three further sporadic third-order sequences, s_7, s_{10}, s_{18} , satisfying recurrences of the form*

$$(n+1)^3 s(n+1) = (2n+1)(an^2 + an + b)s(n) - n(cn^2 + d)s(n-1),$$

with (a, b, c, d) equal to $(13, 4, -27, 3)$, $(6, 2, -64, 4)$, $(14, 6, 192, -12)$ respectively, all integer-valued. Their generating functions are weight-2 modular forms in Hauptmoduln of $\Gamma_0(7)^+$, $\Gamma_0(10)^+$, $\Gamma_0(18)^+$.

The four-parameter recurrence is Gorodetsky’s *template*; the order-3 operator it defines, L_3 , is the object whose symmetric-square structure is the first result of Chapter 8. Let us record here the elementary but central observation that connects the template to Theorem 2.1.

PROPOSITION 3.4 ((K); the template is a symmetric square). *For every $(a, b, c, d) \in \mathbb{Q}^4$, the θ -form operator L_3 of the template above factors as $L_3 = P_2 \cdot \text{Sym}^2(L_2)$ for an explicitly determined second-order operator L_2 whose coefficients are polynomials in z with coefficients in $\mathbb{Q}[a, b, c, d]$, and the leading coefficient P_2 of L_2 satisfies $\theta(P_2) = 2P_1$ where P_1 is the next coefficient. The statement is four polynomial identities, each closed by `ring`.*

The proof is in `Agora/Sequences/PartnerOperators.lean`; what it proves, and what its statement does *not* include, is discussed in Chapters 8 and 9. We only note here that the existence of a symmetric-square structure for these three operators is *consistent with* Doran’s theorem, and that consistency is not confirmation (Warning 2.3): the template is a symmetric square for all (a, b, c, d) , including the uncountably many parameter values that correspond to no K3 family at all.

4. Monodromy and the transcendental lattice

The monodromy of L acts on its solution space, i.e. on $T \otimes \mathbb{C}$, and preserves the lattice T and its form. Conversely, given only the operator, one can hope to *recover* T : compute the monodromy matrices around the singular points numerically, find the invariant symmetric bilinear form, and find the lattice generated by the orbit of a suitable vector. This is a program rather than a theorem, and its output is only as good as its inputs: the monodromy is computed numerically (N), the recognition of its entries as rationals is a recognition step, and the identification of the resulting rank-3 lattice with $T(X_z)$ requires knowing that the family is the one the operator came from. Chapter 8 reports one such computation with every step labelled.

5. Exercises

EXERCISE 5.1. Compute Sym^2 of $\theta^2 - z(\theta + \frac{1}{2})^2$ (the hypergeometric operator of $K(k)$) and identify the result as the Picard–Fuchs operator of the Legendre family’s symmetric square.

EXERCISE 5.2 ($\star$). Show that if y_1, y_2 solve $\theta^2 y + p_1 \theta y + p_0 y = 0$, then $u = y_1 y_2$ satisfies a third-order equation, and that $y_1^2, y_1 y_2, y_2^2$ satisfy the quadratic relation $(y_1 y_2)^2 = y_1^2 \cdot y_2^2$. This is the “nondegenerate quadric” of Doran’s proof.

EXERCISE 5.3. Verify from Theorem 3.3 that $s_7(1) = 4$, $s_7(2) = 48$, $s_7(3) = 760$ (with $s_7(0) = 1$), and that $4 \mid s_7(n)$ for $1 \leq n \leq 10$. (The general statement $4 \mid s_7(n)$ for all $n \geq 1$ is kernel-checked in the program, `four_dvd_s7`.)

Chapter note

Doran’s theorem was read from a hash-pinned copy of [Dor00] (its Thm. 5.13, Cor. 5.8, and §6). Zagier’s and Cooper’s classifications are quoted from the cited papers; the parameter quadruples are those encoded and golden-tested in the program’s Lean development against the first terms of the sequences as printed in the sources.

CHAPTER 6

Singular K3 surfaces and complex multiplication

Where the Picard number of a one-parameter family jumps from 19 to 20, the transcendental lattice drops from rank 3 to rank 2: it becomes a positive-definite even binary quadratic form, and by Shioda–Inose that form determines the surface. In the M_n -polarized families these jumps happen at the CM points of the modular curve $X_0(n)^+$, and the binary form is the orthogonal complement of the algebraic class that appeared. This chapter is where the lattice arithmetic of Chapter 1 pays for itself.

1. Shioda–Inose structures

DEFINITION 1.1 (Morrison [Mor84]). A K3 surface X has a *Shioda–Inose structure* if it carries a symplectic involution ι such that the minimal resolution Y of X/ι is a Kummer surface $\text{Km}(A)$, and the induced rational map $X \dashrightarrow Y$ of degree 2 induces a Hodge isometry $T(X) \cong T(A)$ (with the form on $T(A)$ multiplied by 2 on one side, depending on conventions — see the source).

THEOREM 1.2 ((L), Morrison [Mor84], Thm. 6.3). *A K3 surface X admits a Shioda–Inose structure if and only if there is a primitive embedding $T(X) \hookrightarrow U^3$; equivalently, if $E_8(-1)^2 \hookrightarrow \text{NS}(X)$ primitively. Every K3 surface with $\rho \geq 19$ has one.*

For $\rho = 19$ this is the statement that $T \cong U \oplus \langle 2n \rangle$ embeds in U^3 (Example 3.4 exhibits it for $n = 7$), and it is the lattice-theoretic reason the M_n -polarized surfaces are “sandwiched” between Kummer surfaces of products of isogenous elliptic curves. For $\rho = 20$:

THEOREM 1.3 ((L), Shioda–Inose [SI77]). *Let T be a positive-definite even binary form of discriminant $D > 0$, with associated imaginary quadratic order of discriminant $-D$, and let $\tau_1, \tau_2 \in \mathbb{H}$ be such that $E_{\tau_1} \times E_{\tau_2}$ is the abelian surface with $T(E_{\tau_1} \times E_{\tau_2}) \cong T$ (this exists and is unique up to isomorphism). Then the K3 surface with transcendental lattice T is the Shioda–Inose partner of $E_{\tau_1} \times E_{\tau_2}$: explicitly, the Inose surface $X_{\alpha, \beta}$ of §4 with $J_i = j(\tau_i)/1728$.*

EXAMPLE 1.4. $T = A_2$, $D = 3$: $E_\omega \times E_\omega$, $J_1 = J_2 = 0$, $(\alpha, \beta) = (0, \pm 1)$. $T = \langle 2 \rangle \oplus \langle 2 \rangle$, $D = 4$: $E_i \times E_i$, $J_1 = J_2 = 1$, $(\alpha, \beta) = (1, 0)$ (with $\alpha^3 = 1$; the three cube roots give isomorphic surfaces). These are the X_3 and X_4 of Example 4.3.

2. Rank jumps along a one-parameter family

Let X_τ , $\tau \in \mathbb{H}$, be the M_n -polarized family with period $\omega(\tau) = e - n\tau^2 f + \tau w \in T_n \otimes \mathbb{C}$ (Chapter 3). For $v \in T_n$ the wall of v is $\{\tau : \langle v, \omega(\tau) \rangle = 0\}$; by Lefschetz (1, 1), at such τ the class v is algebraic, so

$$\text{NS}(X_\tau) \supseteq M_n \oplus \mathbb{Z}v \quad (\text{finite index}), \quad \rho(X_\tau) = 20, \quad T(X_\tau) = v^\perp \cap T_n \quad (\text{when } v \text{ is primitive and NS is ex})$$

Since $\langle v, \omega(\tau) \rangle = -nv_0\tau^2 + v_1 + 2nv_2\tau$ is a quadratic in τ with integer coefficients, its roots are quadratic imaginary numbers when $v^2 < 0$: the wall of v consists of *CM points*. The discriminant of the quadratic is $4n^2v_2^2 + 4nv_0v_1 = 2nv^2$, so

$$\tau = \frac{-2nv_2 \pm \sqrt{2nv^2}}{-2nv_0} \quad \text{is a CM point of discriminant } 2nv^2/\gcd^2,$$

and the transcendental lattice at that point is the binary form $v^\perp$, whose discriminant is $\text{disc}(v^\perp) = -v^2 \cdot 2n/\text{div}(v)^2$ by Proposition 2.1. Everything in this paragraph is elementary

except the identification $T(X_\tau) = v^\perp$, which uses that $\text{NS}(X_\tau)$ is exactly the saturation of $M_n \oplus \mathbb{Z}v$ (true for the generic point of the wall by dimension count and Torelli) **(L)**.

EXAMPLE 2.1 (the walls of T_7 , revisited). Take $n = 7$ and the four classes of Chapter 1.

v	v^2	$\text{div } v$	wall τ	$T(X_\tau) = v^\perp$, disc
$(1, -1, 0)$	-2	1	$7\tau^2 = -1$: $\tau = i/\sqrt{7}$	$\langle 2 \rangle \oplus \langle 14 \rangle$, 28
$(2, -4, 1)$	-2	2	$14\tau^2 - 14\tau + 4 = 0$: $\tau = \frac{1}{2} + \frac{i}{2\sqrt{7}}$	$\begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$, 7
$(14, -14, -5)$	-42	14	$\tau = -\frac{5}{14} + \frac{i\sqrt{3}}{14}$	A_2 , 3

The lattice column is kernel-checked **(K)** (Examples 2.2–2.4); the wall column is the elementary quadratic above **(E)**. Note the discriminants 28, 7, 3: by Theorem 4.2 these three points of the M_7 -family are the singular K3 surfaces with those transcendental lattices, and the third is X_3 itself. The CM points have discriminants -28 , -7 , -3 as points of $\mathbb{H}$ — the discriminant of the transcendental lattice and the discriminant of the CM point agree, as they must (both are $2n v^2 / \text{div}(v)^2$ up to sign).

EXAMPLE 2.2 (T_{10}). For $n = 10$: $(1, -1, 0)$ gives $\langle 2 \rangle \oplus \langle 20 \rangle$ (disc 40) at $\tau = i/\sqrt{10}$; $(10, -10, -3)$ gives $\langle 2 \rangle \oplus \langle 2 \rangle$ (disc 4) at $\tau = -\frac{3}{10} + \frac{i}{10}$, i.e. X_4 **(K)** for the lattice (Example 2.5), **(E)** for the wall.

WARNING 2.3 (forced, not corroborating). When a program computes the CM point τ numerically, recognizes it as an element of an imaginary quadratic field of discriminant $-D$, and separately computes the binary form $v^\perp$ and finds its discriminant to be D , it is tempting to report “two independent computations agree”. They are not independent: both are $2n v^2 / \text{div}(v)^2$. The agreement is *forced* by the algebra above and carries no evidential weight about the surface. What *does* carry weight is whether the recognised τ maps, under the family’s parametrization $z(\tau)$, to a point of the z -line that one can check independently — for instance a singular point of the Picard–Fuchs operator. Chapter 8 reports such a check.

3. Which points are ample, and which are not

Dolgachev’s Theorem 2.2(iii) removes the (-2) -walls from the moduli space of *pseudo-ample* polarized surfaces: at the wall of a root δ , the class δ is effective and the polarization class meets it trivially, so the polarization is no longer ample on the surface; the “surface at the wall” is a member with an extra (-2) -curve, or its contraction to a nodal surface. So among the three points of Example 2.1, the first two (roots) are of that nature, while the third (norm -42 , not a root) is a genuine smooth M_7 -polarized surface of Picard number 20 — and it is the one with $T = A_2$, of smallest possible discriminant. This is not a coincidence of the example but a general pattern: the smallest-discriminant CM point of $X_0(n)^+$ is never on a (-2) -wall when $n > 1$, since a root’s complement has discriminant $2n / \text{div}^2 \geq 2n/4$.

4. Exercises

EXERCISE 4.1 ($\star$). Derive the discriminant $2n v^2$ of the wall quadratic and the formula $\text{disc}(v^\perp) = -2n v^2 / \text{div}(v)^2$, and conclude that the CM discriminant and the form discriminant always agree.

EXERCISE 4.2. For $n = 7$, enumerate all primitive $v \in T_7$ with $-v^2 \leq 44$ up to the action of the swap and translation, compute $\text{disc}(v^\perp)$, and list the distinct binary forms that occur. Which of them are on (-2) -walls?

EXERCISE 4.3. Show that the wall equations of $(2, -4, 1)$ and $(-2, 4, 1)$ in T_7 have roots differing by exactly 1, so that the two roots define the same point of $X_0(7)^+$ although they are different vectors with different (isometric) complements.

Chapter note

Theorems of Morrison [[Mor84](#)] and Shioda–Inose [[SI77](#)] are quoted; the reader should check the normalization of the Hodge isometry in Morrison’s definition against the source. The tables of walls are elementary computations reproduced for this book; the lattice halves are kernel-checked as stated. The observation about ampleness follows Dolgachev’s Thm. 7.3 and the inequality on discriminants is immediate from Proposition [2.1](#).

CHAPTER 7

Modular curves and the Fricke involution

The moduli curve of M_n -polarized K3 surfaces is the Fricke modular curve $X_0(n)^+$. This chapter collects what a reader needs about $\Gamma_0(n)$, the Fricke and Atkin–Lehner involutions, and Hauptmoduln, and then shows the correspondence with the lattice T_n concretely enough that the reader can pass back and forth between a point of $X_0(7)^+$ and a class in $U \oplus \langle 14 \rangle$ without consulting a reference.

1. $\Gamma_0(n)$ and its extensions

DEFINITION 1.1. $\Gamma_0(n) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : n \mid c \right\}$. The *Fricke involution* is $W_n = \begin{pmatrix} 0 & -1 \\ n & 0 \end{pmatrix}$, acting on $\mathbb{H}$ by $\tau \mapsto -1/(n\tau)$; it normalizes $\Gamma_0(n)$ and $W_n^2 \in \mathbb{Z}^\times \cdot \Gamma_0(n)$. The group $\Gamma_0(n)^+ = \langle \Gamma_0(n), W_n \rangle$ has index 2 over $\Gamma_0(n)$. More generally, for each $Q \parallel n$ (i.e. $Q \mid n$, $\gcd(Q, n/Q) = 1$) the *Atkin–Lehner involution* is $W_Q = \frac{1}{\sqrt{Q}} \begin{pmatrix} Qa & b \\ nc & Qd \end{pmatrix}$ with $Qad - (n/Q)bc = 1$; the W_Q generate a group $(\mathbb{Z}/2)^{\omega(n)}$ of involutions modulo $\Gamma_0(n)$, and $\Gamma_0(n)^*$ is the extension by all of them [AL70, §5], [DS05, Ch. 3].

The modular curves $X_0(n) = \Gamma_0(n) \backslash \mathbb{H}^*$ and $X_0(n)^+ = \Gamma_0(n)^+ \backslash \mathbb{H}^*$ are compact Riemann surfaces. $X_0(7)$ has genus 0; so do $X_0(7)^+$, $X_0(10)^+$ and $X_0(18)^+$, the three curves of Theorem 3.3.

2. Fixed points and CM points

PROPOSITION 2.1 (**(L)**; elementary). *The fixed points of W_n on $\mathbb{H}$ are the solutions of $n\tau^2 = -1$, i.e. $\tau = i/\sqrt{n}$, together with their $\Gamma_0(n)$ -translates; on $X_0(n)^+$ the fixed points of the involution $X_0(n) \rightarrow X_0(n)^+$ are the points of $X_0(n)$ represented by τ with $\tau = \gamma(-1/(n\tau))$ for some $\gamma \in \Gamma_0(n)$, and their number is a class-number expression: for $n = 7$ there are two, of discriminants -28 and -7 , since $h(-28) + h(-7) = 1 + 1 = 2$ [Fri11, Ogg74].*

The two fixed points are, in the lattice, the walls of the two roots $e - f$ (discriminant -28) and $(2, -4, 1)$ (discriminant -7) of Example 2.1 — and by Theorem 2.2 they are exactly the two points removed from the moduli space of pseudo-ample M_7 -polarized surfaces. The count is a first sanity check of the whole correspondence: two roots up to the group, two fixed points, two singular points of the Picard–Fuchs operator of Chapter 8.

3. Hauptmoduln

DEFINITION 3.1. If $X_0(n)^+$ has genus 0, a *Hauptmodul* is a generator t of its function field, unique up to Möbius transformations; it is *normalized* by prescribing its expansion at the cusp, $t = q^{-1} + O(q)$ or $t = q + O(q^2)$ with $q = e^{2\pi i \tau}$. Dolgachev’s Theorem 3.1(iv) says K_{M_n} has a canonical integer-coefficient Hauptmodul.

EXAMPLE 3.2 ($n = 7$). $h(\tau) = (\eta(7\tau)/\eta(\tau))^4$ is a Hauptmodul for $\Gamma_0(7)$, with $h = q + 4q^2 + \dots$; under the Fricke involution $h(-1/(7\tau)) = 1/(49h(\tau))$ by the transformation law of η (**(L)**). Hence any rational function of h invariant under $h \mapsto 1/(49h)$ descends to $X_0(7)^+$; the degree-2 one

$$z(h) = \frac{h}{1 + 13h + 49h^2}$$

is a Hauptmodul for $X_0(7)^+$ (it has degree 2 in h and is W_7 -invariant, so it has degree 1 on the quotient) **(E)** for the invariance $z(1/(49h)) = z(h)$ **(K)** (`z0f_fricke`). Its values at the two fixed points $h = \pm \frac{1}{7}$ are $z = \frac{1}{27}$ and $z = -1$ **(K)**.

The reader should pause on the number 13: it is the parameter a of the s_7 recurrence (Theorem 3.3), and $49 = 7^2$. The leading coefficient of the s_7 partner operator is $P_2 = 1 - 26z - 27z^2 = (1+z)(1-27z)$, vanishing exactly at $z = -1$ and $z = \frac{1}{27}$; pulled back to the h -line, $P_2(z(h)) \cdot (1 + 13h + 49h^2)^2 = (1 - 49h^2)^2$ **(K)** (`s7_p2_discriminant`): a perfect square vanishing at the two Fricke fixed points. That is the sharp form of the statement “the singular points of the operator are the elliptic points of the modular curve”.

4. The dictionary

modular curve $X_0(n)^+$	lattice $T_n = U \oplus \langle 2n \rangle$	K3 surface
$\tau \in \mathbb{H}$	isotropic $\omega(\tau) = (1, -n\tau^2, \tau)$	period of X_τ
$\Gamma_0(n)$	$\rho(\Gamma_0(n)) \subset O^+(T_n)$	automorphisms of the family
Fricke W_n	swap $e \leftrightarrow f = \text{reflection in } e - f$	—
W_n -fixed point $i/\sqrt{n}$	wall of $e - f$	$\rho = 20$, extra (-2) -curve
CM point of disc. $-D$	v with $2nv^2/\text{div}^2 = -D$	$\rho = 20$, $T = v^\perp$, disc D
cuspid	—	degenerate fibre of the family
Hauptmodul z	—	parameter of the Picard–Fuchs equation

Each row of the dictionary has been met in an earlier chapter; the point of putting them side by side is that the middle column is where computations are *finite*. A question about a CM point on a modular curve becomes a question about an integer vector in a 3×3 Gram matrix, and those are the questions a proof assistant answers without complaint.

5. Exercises

EXERCISE 5.1. Show that W_n normalizes $\Gamma_0(n)$ and that $W_n^2 = -n \cdot I$, so that W_n acts as an involution on $\mathbb{H}$.

EXERCISE 5.2 ($\star$). Verify $z(1/(49h)) = z(h)$ for $z(h) = h/(1 + 13h + 49h^2)$ by clearing denominators, and compute $z(\pm \frac{1}{7})$.

EXERCISE 5.3. Using the q -expansion $h = q + 4q^2 + 14q^3 + \dots$, compute $z(h(q))$ to $O(q^4)$ and compare with the generating function $\sum s_7(n)z^n$ inverted — or rather, check to $O(q^4)$ that $\sum_n s_7(n)z(q)^n$ is the square of a theta series $\sum_{a,b} q^{a^2+ab+2b^2}$. (The program checks this to $O(q^{40})$; it is evidence, not proof, and is the “independent check” of Warning 2.3.)

Chapter note

The theory of $\Gamma_0(n)$ and Atkin–Lehner involutions is standard [DS05, AL70]; the fixed-point count is classical (Fricke, Ogg). The η -quotient Hauptmodul for level 7 and its Fricke behaviour are standard; the specific rational function $z(h)$ and its properties are as verified in the program (kernel-checked where marked, exact q -expansion check to order 40 otherwise).

CHAPTER 8

The frontier: Cooper’s sporadic operators and the M_7 -polarized family

This chapter assembles, with every claim marked, what is currently known about the third-order operators of Cooper’s sporadic sequences and their relation to the M_n -polarized K3 families. It is a status report rather than a theory, and it is included because a textbook that stops where the textbooks stop leaves the reader without a picture of what an open problem in this subject looks like. The detailed account, with proofs and certificates, is the research paper [Cal26b]; here we give the structure and the marks.

1. The template theorem

THEOREM 1.1 ((K); [Cal26b], Thm. 4.1). *For every $(a, b, c, d) \in \mathbb{Q}^4$, the order-3 θ -operator L_3 of the Cooper–Gorodetsky template*

$$(n+1)^3 s(n+1) = (2n+1)(an^2 + an + b)s(n) - n(cn^2 + d)s(n-1)$$

satisfies $L_3 = P_2 \cdot \text{Sym}^2(L_2)$ with $L_2 = P_2\theta^2 + P_1\theta + P_0$ explicitly given by polynomials in z over $\mathbb{Q}[a, b, c, d]$, and $\theta(P_2) = 2P_1$. The Almkvist–van Straten self-adjointness criterion $W \equiv 0$ holds identically in the parameters.

The uniformity is the point: the literature treats the sporadic operators one at a time, and Doran’s theorem gives *existence* of a symmetric-square structure under a geometric hypothesis; here the partner is *exhibited* from the template parameters with no hypothesis at all. (Warning 2.3 applies with full force.)

2. The arithmetic separation

THEOREM 2.1 ((K) except where marked). *Of the three sporadic sequences, s_7 is the only one whose partner operator L_2 has an integral holomorphic solution: the s_{10} and s_{18} partners fail integrality at $n = 2$ (values $17/2$ and $45/2$) (K); the s_7 partner is the formal square root of $\sum s_7(n)z^n$ (K) (`partner_eq_sqrt_s7`), and its integrality follows from $4 \mid s_7(n)$ for $n \geq 1$ (K) (`four_div_s7`), with no literature input. The modular interpretation of the s_7 partner as a weight-1 form is O’Brien’s [O’B16] (L).*

3. The computed monodromy lattice

Numerical Claim (N) (Stream 2, certificates C2_cooper_s7_v5/v6)

Analytic continuation of $L_2(s_7)$ to 60 digits, symmetric-squared, and rational recognition of the monodromy entries (residuals $\sim 10^{-59}$), followed by an exact rational lattice pipeline, gives a joint monodromy-invariant lattice of rank 3, even, discriminant -14 , signature $(2, 1)$, discriminant form $\mathbb{Z}/14$ with $q = 1/14$; an explicit $\text{GL}_3(\mathbb{Z})$ base change identifies it with $U \oplus \langle 14 \rangle = T_7$. In the current certificate the stage-2 monodromy matrices are *certified* (ball arithmetic with rigorous tail bounds, each entry the unique rational of small denominator in its enclosure); the identification of this lattice with $T(X)$ of the s_7 family remains the framework statement (Dolgachev, Doran) and is not certified.

THEOREM 3.1 ((K); the lattice arithmetic). *Everything in Chapter 1’s worked examples: T_N is even of discriminant $-2N$; the glue $e \pm Nf$; the swap is an isometry realizing the Fricke involution on the period; an explicit embedding witness of $T_7 \oplus (U \oplus \langle -14 \rangle)$ into U^3 with index 14 and the rank-22 assembly with $E_8(-1)^2$; the modular group $\Gamma_0(N)$ and all Atkin–Lehner elements inside $O^+(T_N)$ with the weight-2 automorphy factor; the two singular points of $L_3(s_7)$ as the images of the Fricke fixed points under $z(h)$; and the exact complements of the classes $e - f$, $(2, -4, 1)$, $(14, -14, -5)$ in T_7 and $e - f$, $(10, -10, -3)$ in T_{10} .*

4. Conjecture and evidence

Conjecture (Tier B in the program’s language)

The s_7 family is M_7 -polarized; its very general member has $\rho = 19$ and $T \cong U \oplus \langle 14 \rangle$; the rank-2 forms at the CM points are as in Example 2.1.

The evidence, each piece with its mark:

- The operator is a symmetric square **(K)** — consistent with Doran, not implied by it (Warning 2.3).
- The computed monodromy lattice is T_7 **(N)** $\rightarrow$ **(E)** — consistent with Dolgachev’s $(M_7)^\perp$.
- The singular points $\{-1, \frac{1}{27}\}$ are the Fricke fixed points of $X_0(7)^+$ **(K)** for the rational map, **(E)** (order 40) for the parametrization of the family by h — this is the one check that is *not* forced (Warning 2.3).
- The CM points of discriminant $-28, -7, -3$ map to $z = \frac{1}{27}, -1, \infty$ **(N)** (recognition at 120 digits, re-checked at 200).

WARNING 4.1. What would refute the conjecture: an exact monodromy computation giving a lattice other than T_7 ; a proof that the s_7 family has generic Picard number 20 or ≤ 18 ; or a rank-19 polarization other than M_7 with the same period domain (this last is exactly the gap Doran’s §6 leaves open). Nothing so far points that way, and nothing so far closes the door.

5. Two rank-3 lattices, not one

Remark 3.3 deserves a chapter-level flag here, because it is the one error the program actually made and caught. The symmetric square of SL_2 acts on the rank-3 lattice of binary forms of discriminant $-4N^2$; the period representation ρ acts on T_N of discriminant $-2N$. A working directive once named them interchangeably. The kernel-checked theorem `no_isometry_GON_TN` is the two-line proof that they are not isometric — an isometry preserves the discriminant — and it exists in the development only because the conflation had occurred. A textbook lesson: when two objects carry the same group action, ask what invariant distinguishes them before giving them the same name.

6. Open problems

- (1) Make the monodromy exact (connection formulae for L_2), turning **(N)** into **(E)** throughout.
- (2) Prove or refute the converse of Doran’s theorem for order-3 operators with the Cooper template’s shape; equivalently, classify rank-19 lattice polarizations with period domain $\mathbb{H}/\Gamma_0(n)^+$.
- (3) Exhibit explicit Weierstrass models for the two $\rho = 20$ members of the s_7 family with $T = A_2$ and $T = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$ and compute their Néron–Severi lattices by Chapter 4’s methods; the first is X_3 (Example 4.3), the second is not among Inose’s special members and would be new to write down.
- (4) Formalize Theorem 2.2’s table and the Shioda–Tate formula so that Example 4.3 becomes **(K)**.

- (5) Formalize Proposition 2.1 in general (not just its instances), and Nikulin's uniqueness theorem for rank 3.

Chapter note

Every mark in this chapter is transcribed from the program's ledger at the commit of Chapter 9; the numerical claims are Stream 2's and are cited by certificate name so the reader can find their provenance. The author's own contribution to the mathematics is confined to what is marked (**K**); the conjecture is stated as a conjecture.

CHAPTER 9

What “kernel-checked” means

This chapter explains the mark **(K)**. It is short, because the idea is simple, and it is here because the idea is routinely misunderstood in both directions — as guaranteeing more than it does, and as guaranteeing less.

1. The setting

The statements marked **(K)** in this book live in a Lean 4 development [dMU21, The26] in the public repository `SocrateAI-DualScaleTopologicalUniverseModel-LeanProposal`, at the tagged release `v0.25-paper-rankjump` (main branch commit `ad64ed2`, 29 September 2026). The lattice interface (Gram matrices, the hyperbolic plane, E_8 , reflections, the norm form) is imported from a second public development [Cal26a]; nothing in this book depends on anything else that development contains.

A Lean theorem is a term whose type is the statement; the *kernel* is a small program that checks the term has that type. When it does, the statement is a consequence of the axioms the term uses. The developments here use exactly Lean’s three standard axioms (propositional extensionality, choice, quotient soundness); every theorem cited in this book was checked with `#print axioms` to depend on those alone, with no `sorry` (the placeholder that admits a goal) and no `native_decide` (which trusts the compiler rather than the kernel).

2. What the mark certifies

Exactly this: *the statement, as written in Lean, follows from the three axioms*. Not the docstring, not the name, not the paper’s paraphrase of it. Here is one of the statements this book uses, verbatim:

Kernel-checked (K)

```
theorem root7_perp_iff (x : Fin 3 → ℤ) :  
pairing T7 x root7 = 0 ↔ ∃ a b : ℤ, x = ![a, 2 * a - 7 * b, b]  
with T7 := TN 7, TN N := ![0, 1, 0; 1, 0, 0; 0, 0, 2 * N], root7 := ![2, -4,  
1], and pairing G x y := x ·v (G *v y) (the dot product of x with the matrix–vector  
product G y). The mathematical symbols are rendered here for the printed page; the  
source uses the library’s Unicode notation.
```

Read it against Example 2.3. It says: x is orthogonal to $(2, -4, 1)$ under the Gram matrix of $U \oplus \langle 14 \rangle$ if and only if x has the form $(a, 2a - 7b, b)$. That is the full content of “ $v_2^\perp$ is spanned by $(1, 2, 0)$ and $(0, -7, 1)$ ”, and it is stronger than “ $(1, 2, 0)$ and $(0, -7, 1)$ are orthogonal to v_2 ” — which would be true of any two vectors in the complement and would not pin the complement down.

3. Two ways to certify less than the name says

The first failure mode is *vacuity*: a statement of the form “there exists a polynomial of degree 3 such that...” that is true of any cubic. The development had such statements in an early generation; they were deleted, and the episode is recorded in its escalation log.

The second is subtler and is the reason this chapter exists: a statement that is true, compiles, and proves *less than its name*. The recorded instance: a lemma named for the primitivity of $e + Nf$ in U whose statement was `IsCoprime (1 : ℤ) N` — true for every N in

every commutative ring, mentioning no vector, no basis and no lattice. Nine such instances were found in one audit, none by any automated gate; all were found by reading statements against names. The repaired version states the vector:

Kernel-checked (K)

```
theorem glue_primitive (N : ℤ) {d : ℤ} {w : Fin 2 → ℤ}
(h : ![1, N] = d • w) : IsUnit d
```

The rule the development adopted, and which this book follows in every **(K)** box, is: *every noun in the name must appear in the statement*. A reader auditing a formal development should read the statements, not the names, and not the docstrings.

4. Producer and verifier

For the lattice results of Chapter 1 the gates were run twice. The author’s session ran the build, the `sorry` and axiom audits and a statement-hash lock, and printed the axioms of each theorem. A second team of the program, in a different repository, then re-ran the same gates on the exact commits, recomputed the lattice arithmetic from its own certificate data before trusting either report, and recorded the result against the commit and the SHA-256 of the source file. Independently, the development [Cal26a] stated the same five complements in its own file (DualScaleDyons/RankJump.lean); the two sets of statements were compared and agree — a comparison of statements, not a second verification of either proof, since neither side built the other’s file. We record one detail because it is the kind of thing a reader should know happens: a control value $\langle v_2, v_3 \rangle$ was first written by hand as -42 ; the kernel rejected it; the value that compiles is -154 .

5. What was not checked

The identification of $U \oplus \langle 14 \rangle$ with the transcendental lattice of the s_7 family; Nikulin’s theorems; Dolgachev’s and Doran’s theorems; the Kodaira–Tate table; the Shioda–Tate formula; every q -expansion identity; every numerical recognition. In short: everything marked **(L)**, **(E)** or **(N)** in this book. The kernel checked 3×3 integer matrices and polynomial identities. That is a great deal less than the theory of K3 surfaces and a great deal more than nothing: it is exactly the part of the argument where a slip of the pen — -42 for -154 , I_1 for I_2 , one rank-3 lattice for another — would otherwise have gone unnoticed.

6. Exercises

EXERCISE 6.1. Write the statement of Example 2.4’s complement in the style of `root7_perp_iff`, and check that every noun in the name “`root7Inf_perp_iff`” occurs in it.

EXERCISE 6.2. Find, in any formal development you have access to, a theorem whose name promises more than its statement delivers. (This is easier than it sounds.)

Chapter note

The repository, commit and release are as stated; the two Lean statements are transcribed verbatim from `Agora/Geometry/MnLattice.lean` at that commit. The account of the audit and of the producer/verifier run follows the repository’s lessons-learned file and the receipt briefs exchanged between the program’s teams, all of which are in the repository.

APPENDIX A

Hints for the exercises

Chapter 1: Integral lattices

- 1.1:** $q(1) = 1/2N$ and $q'(1) = -1/2N$ in $\mathbb{Q}/2\mathbb{Z}$ on $\mathbb{Z}/2N$. An isomorphism must send 1 to a unit u with $u^2/2N \equiv -1/2N \pmod{2}$, i.e. $u^2 \equiv -1 \pmod{4N}$. So the forms are isomorphic iff -1 is a square modulo $4N$. For $N \leq 12$ this happens for no N except... check $N = 1$: $4N = 4$, $-1 \equiv 3$ is not a square mod 4. Indeed -1 is never a square modulo 4, so *never*: $\langle 2N \rangle$ and $\langle -2N \rangle$ have non-isomorphic discriminant forms for every N . (The exercise's phrasing invited you to compute before believing.)
- 1.2:** $F^\top GF$ is block-diagonal with blocks (v^2) and the Gram matrix of $v^\perp$; take determinants; $|\det F|$ is the index, which is $|v^2|/\text{div } v$ by Proposition 2.1.
- 1.3:** Roots satisfy $2v_0v_1 + 14v_2^2 = -2$, i.e. $v_0v_1 = -1 - 7v_2^2$. For $v_2 = 0$: $(1, -1, 0), (-1, 1, 0)$. For $v_2 = \pm 1$: $v_0v_1 = -8$, giving $(\pm 1, \mp 8), (\pm 2, \mp 4), (\pm 4, \mp 2), (\pm 8, \mp 1)$ with $v_2 = \pm 1$. $\text{div} = \gcd(v_1, v_0, 14v_2)$: divisor 2 for $(\pm 2, \mp 4, \pm 1)$ and $(\pm 4, \mp 2, \pm 1)$, divisor 1 otherwise. Complements have discriminant 28 (divisor 1) or 7 (divisor 2); reduced forms $x^2 + 7y^2$ and $x^2 + xy + 2y^2$.
- 1.4:** $\langle x, (N, -N, -k) \rangle = -Nx_0 + Nx_1 - 2Nkx_2$; $v^2 = -2N^2 + 2Nk^2 = 2N(k^2 - N)$; $\text{disc } v^\perp = -v^2 \cdot 2N/N^2 = 4(N - k^2)$ when the divisor is exactly N , i.e. $2k \not\equiv 0 \pmod{N}$ conditions aside. $N = 7, k = 5$: $v^2 = -42$, $\text{disc} = \dots$ careful: $\text{disc}(v^\perp) = -v^2 \cdot 2N/\text{div}^2 = 42 \cdot 14/196 = 3$. $N = 10, k = 3$: $v^2 = -20$, $\text{disc} = 20 \cdot 20/100 = 4$.
- 1.5:** P sending $(e, f, w) \mapsto (-f, w, e)$ up to signs; check $P^\top T_N P$ entrywise.
- 1.6:** $\det$ is preserved by $P^\top GP$ with $\det P = \pm 1$.

Chapter 2: K3 surfaces

- 2.1:** For the quartic, $c(T_X) = (1 + h)^4/(1 + 4h)$ restricted; $c_2 = 6h^2 = 24$.
- 2.2:** M_n primitive in NS with $\text{rk NS} = 19$ forces $\text{NS} = M_n$; then $T = \text{NS}^\perp = (M_n)^\perp$ and Example 3.4 applies. Both hypotheses are load-bearing.
- 2.3:** Reduced forms $ax^2 + bxy + cy^2$ with $|b| \leq a \leq c$, $b^2 - 4ac = -D$. $D = 7$: $(1, 1, 2)$ only. $D = 28$: $(1, 0, 7)$ and $(2, 2, 4)$ (non-primitive). $D = 40$: $(1, 0, 10)$ and $(2, 0, 5)$.

Chapter 3: Periods, Torelli, and lattice polarizations

- 3.2:** $\langle (1, -1, 0), (1, -n\tau^2, \tau) \rangle = 1 \cdot (-n\tau^2) + (-1) \cdot 1 + 0 = -(n\tau^2 + 1)$. For $(2, -4, 1)$ in T_7 : $2(-7\tau^2) + (-4) + 14\tau = 0 \iff 14\tau^2 - 14\tau + 4 = 0 \iff \tau = \frac{1}{2} \pm \frac{i}{2\sqrt{7}}$. For $(-2, 4, 1)$: $14\tau^2 + 14\tau + 4 = 0$, roots $-\frac{1}{2} \pm \frac{i}{2\sqrt{7}}$.
- 3.3:** Translation acts on $(1, -n\tau^2, \tau)$ by $\tau \mapsto \tau + 1$; the matrix is $\begin{pmatrix} 1 & 0 & 0 \\ -n & 1 & -2n \\ 1 & 0 & 1 \end{pmatrix}$ (columns act on coordinates $(1, -n\tau^2, \tau)$); check it preserves T_n .

Chapter 4: Elliptic fibrations

- 4.2:** $\Delta = -16 \cdot 27a_6^2 = -432t^{10}(t-1)^4$; orders 10 at 0, 4 at 1, and $24 - 14 = 10$ at ∞ .
- 4.3:** $4(-3t^4)^3 + 27t^{10}(t^2+1)^2 = -108t^{12} + 27t^{10}(t^2+1)^2 = 27t^{10}((t^2+1)^2 - 4t^2) = 27t^{10}(t^2-1)^2$; so $\text{ord}_{\pm 1} = 2$ and $a_4(\pm 1) = -3 \neq 0$.
- 4.4:** Two II^* give E_8^2 , rank 16, plus U : 18; the two missing units of rank come from $\sum(m_v - 1)$ over the remaining four units of Euler number, or from MW.

Chapter 5: Picard–Fuchs equations

5.2: Differentiate $u = y_1 y_2$ three times and eliminate using the second-order equation; the relation is Wr-free.

5.3: $s_7(1) = (1)(4)/1 = 4$; $s_7(2) = (3 \cdot (13 + 13 + 4) \cdot 4 - 1 \cdot (-27 + 3) \cdot 1)/8 = (360 + 24)/8 = 48$;
 $s_7(3) = (5 \cdot (52 + 26 + 4) \cdot 48 - 2 \cdot (-108 + 3) \cdot 4)/27 = (19680 + 840)/27 = 760$.

Chapter 6: Singular K3 surfaces

6.1: The wall quadratic is $-nv_0\tau^2 + 2nv_2\tau + v_1 = 0$, discriminant $4n^2v_2^2 + 4nv_0v_1 = 2n(2v_0v_1 + 2nv_2^2) = 2nv^2$. Divide by $\text{div}(v)^2$ to make it primitive; compare with Proposition 2.1.

6.3: See hint 3.2.

Chapter 7: Modular curves

7.2: $z(1/(49h)) = \frac{1/(49h)}{1 + 13/(49h) + 1/h^2} = \frac{h}{49h^2 + 13h + 1} = z(h)$. $z(\frac{1}{7}) = \frac{1/7}{1 + 13/7 + 1} = \frac{1/7}{27/7} = \frac{1}{27}$; $z(-\frac{1}{7}) = \frac{-1/7}{1 - 13/7 + 1} = \frac{-1/7}{1/7} = -1$.

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